

§ Group Action on a Set

Recall Binary operation

$$\begin{array}{ccc} * : S \times S & \longrightarrow & S \\ \downarrow & & \downarrow \\ (a, b) & \longmapsto & a * b \end{array} \quad \text{where } a * b := * (a, b)$$

Def G : a group, X : a set

$$* : G \times X \longrightarrow X \quad \text{satisfying} \quad \begin{array}{l} \textcircled{1} \quad ex = x \quad \forall x \in X \\ \textcircled{2} \quad (gg_1)x = g(g_1x) \quad \forall g, g_1 \in G, \forall x \in X \end{array} \quad \text{where } gx := g * x = * (g, x).$$

Then $*$ is called a group action of G on X .

We say ' G acts on X ' or ' X is a G -set' and denote $G \curvearrowright X$.

Thm X : G -set

Then $\forall g \in G$, the map $\sigma_g : \begin{array}{ccc} X & \longrightarrow & X \\ x & \longmapsto & gx \end{array}$ is a permutation of X .

pf) (injective) $\sigma_g(x) = \sigma_g(y)$ i.e. $gx = gy \implies g^{-1}gx = g^{-1}gy$

$$\implies (g^{-1}g)x = (g^{-1}g)y \quad \text{by } \textcircled{2}$$

$$\implies ex = ey$$

$$\implies x = y \quad \text{by } \textcircled{1}$$

(surjective) $\forall x \in X, \sigma_g(g^{-1}x) = gg^{-1}x = x$.

Thm $G \curvearrowright X$, $\varphi: G \longrightarrow S_X$ is a homomorphism.

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & S_X \\ g & \mapsto & \sigma_g: X \longrightarrow X \\ & & x \mapsto gx \end{array}$$

pf) $\varphi(gg_2) = \sigma_{gg_2}$, $\varphi(g_1) \circ \varphi(g_2) = \sigma_{g_1} \circ \sigma_{g_2}$.

Let $x \in X$. Then (i) $\sigma_{gg_2}(x) = gg_2x$, (ii) $\sigma_{g_1} \circ \sigma_{g_2}(x) = \sigma_{g_1}(g_2x) = gg_2x$.

Remark G : a group, X : a set

Then there is a 1-to-1 correspondence between

both are not empty

$$\{\text{Actions of } G \text{ on } X\} \longleftrightarrow \{\text{Homomorphisms } f: G \longrightarrow S_X\}$$

Note that $G \curvearrowright X \longmapsto \varphi: G \longrightarrow S_X$

$$\begin{array}{ccc} g * x = \varphi(g)(x) & \longleftrightarrow & \varphi: G \longrightarrow S_X \\ & & \begin{array}{ccc} G & \xrightarrow{\varphi} & S_X \\ g & \mapsto & \sigma_g: X \longrightarrow X \\ & & x \mapsto \sigma_g(x) \end{array} \end{array}$$

check

$$\textcircled{1} e * x = \varphi(e)(x) = I_X(x) = x, \quad \textcircled{2} (gg_2) * x = \varphi(gg_2)(x) = \varphi(g_1) \circ \varphi(g_2)(x) = \varphi(g_1)(g_2x) = g_1 * (g_2 * x).$$

Remark $G \curvearrowright X \xrightarrow{\text{induce}} \varphi: G \longrightarrow S_X$.

$$\text{Then } \ker(\varphi) = \{g \in G : \varphi(g) = I_X\}$$

$$= \{g \in G : gx = x \text{ for all } x \in X\}$$

is a set that precisely fix every element of X by $G \curvearrowright X$.

Def $G \curvearrowright X$

i) We say the action $G \curvearrowright X$ is faithful if $gx = x \iff g = e$ for all $x \in X$.

ii) " " is transitive if $\forall x, x_2 \in X, \exists g \in G$ s.t. $x_2 = gx_1$.

Ex $G = X$, $G \curvearrowright G$ by multiplication ① $e * x = ex = ex$ is faithful and transitive.

$$\textcircled{2} (g, g_2) * x = g(g_2 x)$$

Ex $G \curvearrowright G$ by conjugation $g * x := gxg^{-1}$ ① $e * x = x$ is not necessarily both faithful and transitive.

$$\textcircled{2} (g, g_2) * x = gg_2 x g_2^{-1} g^{-1}$$

$$g * (g_2 * x) = g * (g_2 x g_2^{-1}) = gg_2 x g_2^{-1} g^{-1}$$

If G is abelian, the above action is not faithful and transitive.

Ex $S_n \curvearrowright \{1, 2, \dots, n\}$ is both faithful and transitive.
 $\sigma * i := \sigma(i)$

Ex $H \leq G$, $\mathcal{L}_H := \{\text{all the left cosets of } H \text{ in } G\}$

Then $G \curvearrowright \mathcal{L}_H$ is a group action.
 $g * aH := (ga)H$

Def $G \curvearrowright X$, $x \in X$, $g \in G$

Define $G_x := \{a \in G : ax = x\}$ and $X_g := \{y \in X : gy = y\} \subseteq X$
not empty *possibly empty*

Simply, G_x is the set of elements in G that fix $x \in X$ and

X_g is the set of elements in X fixed by $g \in G$.

Ex $G = S_3$, $X = \{1, 2, 3\}$, $g_1 = (1, 2)$, $g_2 = (1, 2, 3)$, $x_1 = 3$

Then $G_{x_1} = \{\sigma \in S_3 : \sigma(3) = 3\} = \{I_3, (1, 2)\} \leq G$

$$X_{g_1} = \{y \in X : g_1(y) = y\} = \{3\}, \quad X_{g_2} = \emptyset.$$

Thm $G \curvearrowright X$. Then $G_x \leq G$ for all $x \in X$.

pf) First, G_x is not empty since $e \in G_x$.

$$\text{ETS: } a, b \in G_x \implies ab^{-1} \in G_x$$

Let $a, b \in G_x$. Then $ax = bx = x$

$$\text{Note that } b^{-1}ax = x = b^{-1}x \implies ax = ab^{-1}x = x$$

Hence, $ab^{-1} \in G_x$.

Note G_x is called an isometry subgroup of x or stabilizer of x .

Remark $X : G\text{-set}$, $x, y \in X$

Define a relation $\sim$ on X s.t. $x \sim y$ if $\exists g \in G$ s.t. $y = gx$.

Then $\sim$ is an equivalence relation.

(reflexive) $x \sim x$ since $x = ex$

$$\text{(symmetric)} \quad x \sim y \implies y = gx \text{ for some } g \in G \implies g^{-1}y = x \implies y \sim x$$

$$\text{(transitive)} \quad x \sim y, y \sim z \implies y = g_1x, z = g_2y \text{ for some } g_1, g_2 \in G \implies z = (g_2g_1)x \implies x \sim z$$

An orbit of $G \curvearrowright X$ is an equivalence class of $\sim$ on X .

For $x \in X$, the orbit of $G \curvearrowright X$ containing x is denoted by Gx . ($Gx = \{y \in X : y \sim x\} = \{gx \in X : g \in G\}$)

$$Gx = Gy \iff x \text{ and } y \text{ are contained in the same orbit}$$

$$\iff y = gx \text{ for some } g \in G.$$

Remark $G \curvearrowright X$ transitively if there is only one orbit.

That is. $Gx = X$ for all $x \in X$.

Thm Orbit Stabilizer Theorem

$X : G\text{-set}, x \in X$

$$\text{Then } |Gx| = [G : G_x] = \frac{|G|}{|G_x|} \quad \begin{array}{l} \nearrow \\ G : \text{finite} \end{array}$$

pf) Define $\mathcal{L}_x$ be the set of all the left cosets of G_x in G .

Consider the map $\psi : \mathcal{L}_x \longrightarrow Gx$ by $\psi(aG_x) = ax$.

(well-defined) WTS: $aG_x = bG_x \implies ax = bx$

Assume $aG_x = bG_x$. Then $a^{-1}b \in G_x$.

$$a^{-1}b \in G_x \implies a^{-1}bx = x \implies bx = ax$$

$$(\text{1-to-1}) \quad \psi(aG_x) = \psi(bG_x) \implies ax = bx$$

$$\implies x = a^{-1}bx$$

$$\implies G_x = a^{-1}bG_x$$

$$\implies aG_x = bG_x$$

(onto) Clear by def

Hence ψ is a bijection so $|\mathcal{L}_x| = |Gx|$.

Thm Burnside's formula

$G \curvearrowright X$, G, X : finite

The number of orbits of $G \curvearrowright X$ is equal to $\frac{1}{|G|} \sum_{g \in G} |X_g|$.

pf) Let $O_1, O_2, \dots, O_k$ be orbits of $G \curvearrowright X$.

Then $X = \bigcup_{i=1}^k O_i$ and $O_i \cap O_j = \emptyset$ for $i \neq j$.

$gx=x$ 인 x 들의 집합을 각각의 g 에 대하여 생각 $gx=x$ 인 g 들의 집합을 각각의 x 에 대하여 생각

Note that $\sum_{g \in G} |X_g| = \sum_{x \in X} |G_x| = |\{(g, x) \in G \times X : gx = x\}|$

$$\text{Orbit-Stabiliser thm} \rightarrow \sum_{x \in X} \frac{|G|}{|G_x|}$$

$$= \sum_{i=1}^k \sum_{x \in O_i} \frac{|G|}{|G_x|}$$

$$x \in O_i \implies G_x = O_i \rightarrow \sum_{i=1}^k \sum_{x \in O_i} \frac{|G|}{|O_i|}$$

$$= \sum_{i=1}^k \frac{|G|}{|O_i|} |O_i|$$

$$= k |G|$$

Hence, the number of orbits of $G \curvearrowright X$ is $k = \frac{1}{|G|} \sum_{g \in G} |X_g|$.